Observation of γ’ precipitate splitting associated with dendritic instability in Ni–Al–X alloys

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Poster No: ICME_075

Introduction

- Size and distribution of precipitates play a key role in determining the superalloys ‘ high-temperature properties.
- Precipitate interaction by γ/γ′ lattice misfit and degree of anisotropy in elastic modulus will dictate the morphology and topological alignment of γ′ precipitates.
- To understand the morphological evolution of γ′ precipitates, time-dependent growth experiments were conducted for the model alloy Ni-10Al-4Ta.

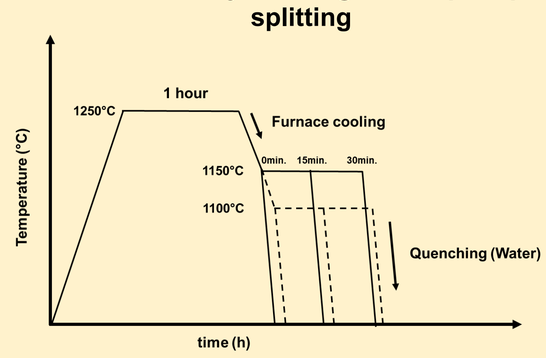
Experimental details

Alloy systems: Ni-10Al-4Ta

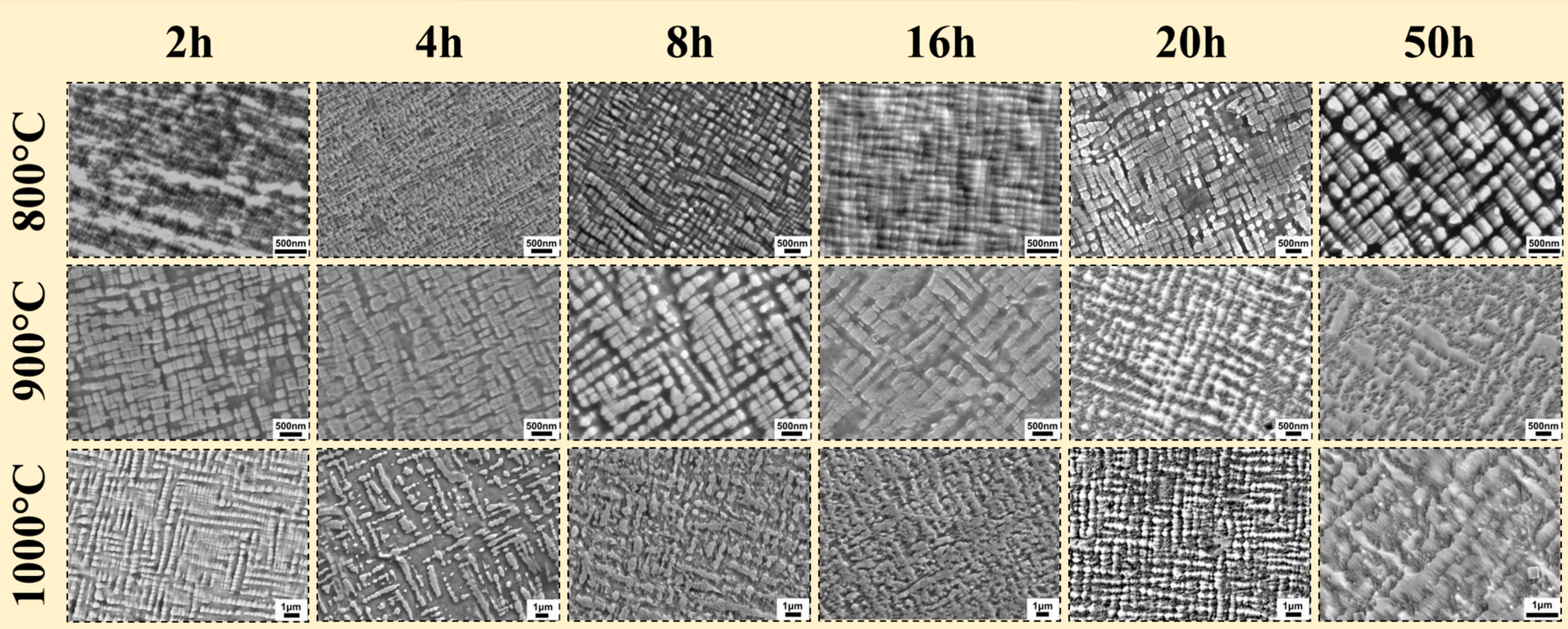
- The alloy was prepared by vacuum arc melting followed by casting into a rod.
- Homogenization of the cast rods at 1350°C for 75h
- Kalling’s reagent No-2 is used for chemical etching
- Growth study is conducted at 800, 900 and 1000°C for 2h, 4h, 8h, 16h, 20h, and 50h.



Heat treatment cycle designed for precipitate splitting

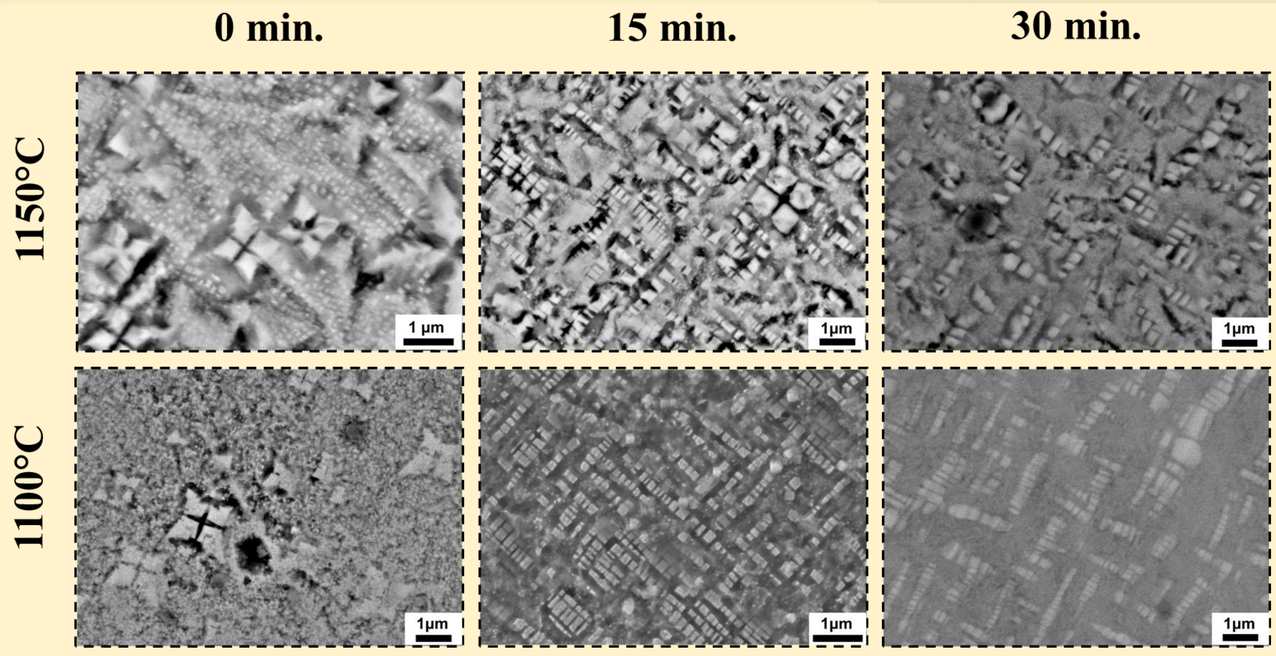


Precipitate morphological evolution at 800, 900 and 1000°C

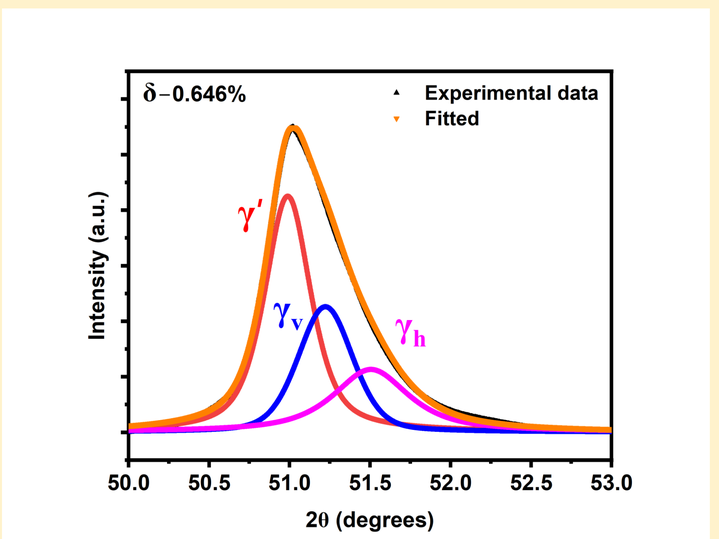


- Precipitate shape is deviating from cubic after 20h, 16h, and 2h at 800, 900 and 1000, isothermal heat treatment respectively.
- The dendritic morphology of precipitates observed at 20h at 900 and 1000C isothermal heat treatment.
- The special heat treatment cycle is designed to witness the splitting of precipitates with dendritic instability as pointed out in [3].

Splitting of Precipitates with Dendritic Instability



γ/γ′ lattice misfit calculations

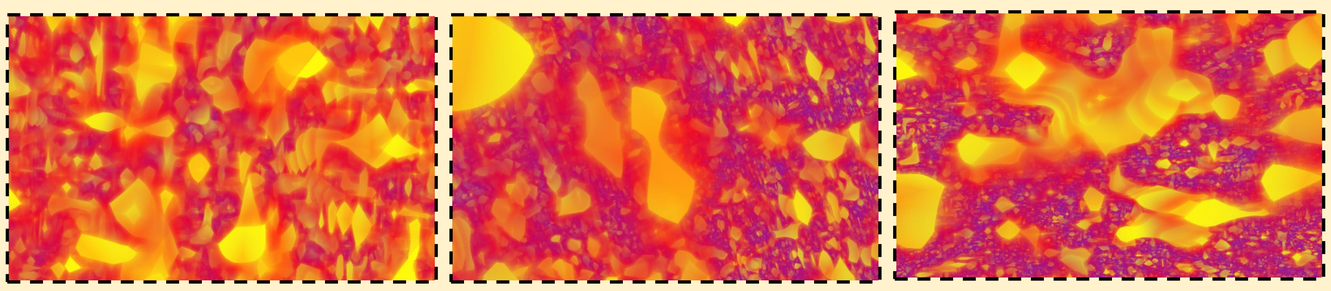


$$a = \frac{2(a_\gamma - a_\gamma)}{(a_\gamma + a_\gamma)}$$

$$a_{\gamma,cubic} = \frac{(S_{11} + S_{12})a_{\gamma,h} - 2S_{12}a_{\gamma,v}}{S_{11} + S_{12}}$$

Where S_{11} and S_{12} represent the elastic compliance values in γ matrix while $a_{\gamma,h}$ and $a_{\gamma,v}$ are lattice parameters in horizontal and vertical channels respectively.

Xray 3D Tomography



- Microstructure consists of primary and secondary γ′ precipitates and PFZ in the vicinity of primary γ′ precipitates assembly[2].
- Primary precipitates are growing as dendrites along the <111> direction[2].
- Tetragonal distortion in the γ-matrix due to lattice misfit constraint.
- This clearly indicates the high degree of anisotropy in the lattice elastic modulus of the γ-matrix.
- Precipitate splitting along the corners is confirmed by the 3D-X-Ray tomography of the precipitates in condition of 0min, 1150C.

Conclusions

- The present alloy system exhibits a precipitate shape instability i.e., splitting and growing into dendrites due to **point effect of diffusion**.
- High γ/γ′ lattice misfit and high degree of elastic anisotropy could be the reason for this splitting.

References

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Acknowledgment

Author would like to acknowledge the department of Materials Engineering, IISc and Advanced Facility For Microscopy and Micro-analysis (AFMM, IISc) for availing the microscope and other characterization facilities.